Skills Summary

Classifying Quadrilaterals from Defining Properties

Use this reference while completing your worksheet or when studying.

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1. Read the marks, then name the most specific shape

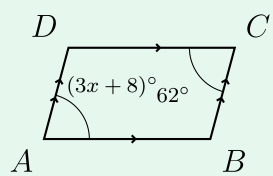
Definitions (each one adds a condition to the one above):

- **Trapezoid** — at least one pair of parallel sides.
- **Parallelogram** — *both* pairs of opposite sides parallel.
- **Rhombus** — a parallelogram with four congruent *sides*.
- **Rectangle** — a parallelogram with four right *angles*.
- **Square** — both at once: four congruent sides *and* four right angles.

Parallelogram angle facts: opposite angles are congruent; consecutive angles are supplementary. Every quadrilateral's angles sum to 360°.

(!) Name what the marks *force*, not what the drawing resembles, and always give the *most specific* name that the marks reach.

Example: Name $ABCD$, then find x.



Step 1. Two pairs of chevrons and no side or angle marks — that is the parallelogram definition and nothing more, so use a *parallelogram* angle fact.

Step 2. $\angle A$ and $\angle C$ are opposite, so they are congruent: $3x + 8 = 62$, giving $3x = 54$.

Parallelogram, and $\boldsymbol{x = 18}$

Try it: In parallelogram $PQRS$, $\angle P$ and $\angle Q$ are consecutive, with $m\angle P = (2x)°$ and $m\angle Q = (x + 30)°$. Find x.

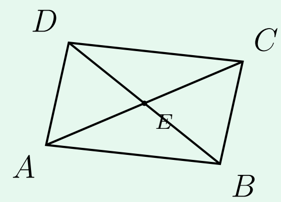
check: $\boldsymbol{x = 50}$

2. Classify from what the diagonals do

Diagonal tests — each one names a family on its own:

- Diagonals **bisect each other** ⇒ parallelogram.
- … and are also **congruent** ⇒ rectangle.
- … and are also **perpendicular** ⇒ rhombus.
- Congruent *and* perpendicular *and* bisecting ⇒ square.

Example: The diagonals of $ABCD$ bisect each other at E, with $AE = 3x - 4$ and $EC = x + 6$. Find x.



Step 1. Bisecting means E is the midpoint of $\overline{AC}$, so the two halves of that diagonal are equal — set them equal rather than adding them.
Step 2. $3x - 4 = x + 6$, so $2x = 10$.
$\boldsymbol{x = 5}$, and the bisecting diagonals make $ABCD$ a parallelogram

Try it: The diagonals of rectangle $PQRS$ are congruent, with $PR = 4x + 1$ and $QS = 2x + 15$. Find x.

check: $\boldsymbol{x = 7}$

3. Classify from vertex coordinates

Slope decides parallel and perpendicular; **distance** decides congruent.

Equal slopes $\Rightarrow$ parallel. Slopes multiplying to $-1 \Rightarrow$ perpendicular, so a right angle.

Both pairs of opposite sides parallel $\Rightarrow$ parallelogram; add a right angle for a rectangle, add equal adjacent sides for a rhombus, add both for a square.

Example: $W(0, 0)$, $X(4, 2)$ and $Y(6, -2)$ are three vertices. Is the angle at X a right angle?
Step 1. A right angle at X is a claim about the two sides meeting there, so compare the slopes of $\overline{WX}$ and $\overline{XY}$.
Step 2. Slope of $\overline{WX}$ is $\dfrac{2-0}{4-0}$, and slope of $\overline{XY}$ is $\dfrac{-2-2}{6-4}$.
$\boldsymbol{\frac{1}{2}}$ and $\boldsymbol{-2}$ multiply to -1, so yes — the angle at X is right

Try it: Find the slope of the segment joining $(2, -1)$ and $(4, 3)$.

check: **2**